

Technical Audit Proposal: Jigsaw Unintended Bias in Toxicity Classification

Ian Tang Nick Chen (Group No.21)

March 2026

1 Background

In this project we audit an Algorithmic Decision-Support System (ADS) used for detecting toxic comments in online discussions. The system we analyze comes from the Kaggle competition *Jigsaw Unintended Bias in Toxicity Classification*.

We selected this ADS because automated content moderation systems are widely used on online platforms and can influence how people communicate online. This task is closely related to course topics such as fairness and bias, because toxicity detection models may incorrectly assign higher toxicity scores to comments mentioning certain identity groups.

The goal of this system is to classify whether a comment is toxic. However, models trained on text data may learn unintended correlations between identity-related words (such as religion, gender, or ethnicity) and toxicity labels. This creates a potential trade-off between predictive accuracy and fairness.

2 Input and Output

The dataset consists of public comments collected from the Civil Comments platform between 2015 and 2017. In our implementation, the training dataset has shape (1,804,874, 45) and the test dataset has shape (97,320, 2).

The data is split into:

- Training set: 1,604,874 comments
- Validation set: 200,000 comments
- Test set: 97,320 comments

The main input feature is the raw comment text. The dataset also contains identity-related attributes, such as gender or religion indicators. These attributes allow us to analyze potential bias in the model predictions.

The output of the system is a toxicity score. This score represents the probability that a comment is toxic. Higher values indicate a greater likelihood of toxic language.

3 Implementation and Preliminary Performance

We reimplemented the 3rd place solution from the Jigsaw Unintended Bias in Toxicity Classification competition, based on team F.H.S.D.Y.’s official repository ([sakami0000/kaggle_jigsaw](https://github.com/sakami0000/kaggle_jigsaw)). The current recreation pipeline runs end-to-end in Google Colab with Drive-cached data and checkpoints for reproducibility across sessions.

The implementation replicates the original solution’s core techniques: BERT-large-uncased with head+tail truncation, metric-aligned sample weighting, multi-task loss with 6 auxiliary targets, negative downsampling, and an LSTM-GRU baseline with Beta-smoothed annotator-count loss. Predictions are blended via Optuna with a robustness filter. We evaluated against the private leaderboard split using `test_private_expanded.csv` and scored **0.94466**, within 0.002 of the original 0.94683. The gap stems from this being a two-model recreation of the framing and techniques, not the full ensemble.

Per-metric breakdown shows Mean Subgroup AUC (0.90183) as the weakest component—the metric most directly tied to identity-group fairness—while Overall AUC (0.97369) is strong. This confirms that the model’s fairness bottleneck lies in subgroup performance, which will be the primary focus of our audit. The full ensemble will be included in the final report to compare duo-model versus ensemble performance on subgroup fairness metrics.

4 Audit Plan

In the next stage of the project, we plan to audit the system in several ways.

First, we will analyze model accuracy across different subgroups using metrics such as ROC-AUC and F1-score. This will help us determine whether the model performs differently across demographic groups.

Second, we will analyze fairness using bias-related metrics such as subgroup AUC and BPSN/BNSP AUC. These metrics help measure whether the model incorrectly assigns higher toxicity scores to comments mentioning certain identities.

Third, we will test robustness by modifying input sentences and observing how the predictions change. For example, we will compare sentences that differ only by identity-related words.

Our audit methodology draws on the SMACTR framework proposed by Raji et al. [3], which defines an end-to-end internal algorithmic auditing process structured around scoping, testing, and reflection stages. We scope our audit to the subgroup fairness properties of a toxicity classification system, test by disaggregating performance across identity-mentioning subgroups using the competition’s bias-aware

metric (a power-mean composite of Subgroup AUC, BPSN AUC, and BNSP AUC), and reflect on how architectural decisions—specifically, ensemble diversity versus single-model reliance—affect fairness outcomes. Following the model reporting recommendations of Mitchell et al. [4], we report per-subgroup evaluation results to surface disparities that aggregate metrics obscure.

5 Early Observation

Our initial exploration suggests that some sentences containing identity-related terms may receive higher toxicity scores even when the content is neutral. This motivates further investigation into potential bias in the system.

Data and Code Sources

The dataset used in this project comes from the Kaggle competition *Jigsaw Unintended Bias in Toxicity Classification* [1]. The implementation of the ADS is based on a publicly available Kaggle solution that reproduces a top-performing model for this competition [2].

References

- [1] Kaggle. *Jigsaw Unintended Bias in Toxicity Classification*. <https://www.kaggle.com/competitions/jigsaw-unintended-bias-in-toxicity-classification>
- [2] Sakami, T. *Kaggle Jigsaw Unintended Bias Solution (3rd place implementation)*. https://github.com/sakami0000/kaggle_jigsaw
- [3] I. D. Raji, A. Smart, R. N. White, M. Mitchell, T. Gebru, B. Hutchinson, J. Smith-Loud, D. Theron, and P. Barnes, “Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing,” in *Proc. ACM FAccT*, 2020.
- [4] M. Mitchell, S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji, and T. Gebru, “Model cards for model reporting,” in *Proc. ACM FAT**, 2019.

Technical Audit: Jigsaw Unintended Bias in Toxicity Classification

DS-UA 202: Responsible Data Science

Ian Tang & Nick Chen (Group No.21)

March 2026

1. Background & Motivation

- **System:** Algorithmic Decision-Support System (ADS) for moderating online public comments.
- **Context:** Kaggle follow-up competition to a previous toxic comment classification challenge.
- **Purpose:** Identify "toxic" content to improve the quality of online conversation.
- **Problem:** Machine learning models often learn spurious correlations between key gender/racial-ethnic/religious identity terms and toxicity.
- **Goal:** Audit the trade-off between toxicity detection accuracy and unintended bias against protected subgroups.

2. Input and Output

- **Data:** 2 million public comments from the Civil Comments platform (2015–2017).
- **Input Features:** Raw `comment_text` with labels for 24 identity-related attributes.
- **Output:** A continuous score representing the fraction of human raters who labeled the text as toxic.
- **Analysis Plan:** Profile feature correlations and value distributions to detect inherent data imbalances in the training set.

3. Implementation & Analysis Plan

- **ADS Implementation:** Audit of a top-tier Kaggle solution; evaluating pre-processing and model architecture.
- **Accuracy Metrics:** Disaggregated ROC-AUC and F1-scores across demographic slices.
- **Fairness Metrics:** Primary focus on **Bias-AUC** (BPSN/BNSP) to measure unintended associations.
- **Additional Audit:** Stability testing via counterfactual fairness (perturbing identity tokens).

4. Preliminary Takeaway & Next Steps

- **Early Finding:** Neutral sentences containing specific identity keywords receive higher toxicity probabilities than identical sentences without them.
- **Evaluation:** Assessing if the model meets the necessary fairness criteria for public sector or industry deployment.
- **Methodology:** Using a Google Colab environment to reproduce the implementation and execute the bias audit.
- **Stakeholder Impact:** Reflecting on the ethical implications for users belonging to the over-flagged subgroups. Mixed impacts on subtype groups in regards to censoring and flagging.